**LABSHEET-6 **

Setup — Google Colab

```
# Shiv(55)
!pip install pillow numpy matplotlib
```

```
Requirement already satisfied: pillow in /usr/local/lib/python3.13/dist-packages (11.3.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (4.66.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (26.3)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.3)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (2.9.0.post
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.7->matplotlib) (
```

Q1. Read an Image and Display Its Properties

```
# Shiv(55)
from google.colab import files
from PIL import Image
import os
import numpy as np

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename)
arr = np.array(img)

print("File Name:", filename)
print("Image Format:", img.format)
print("Dimensions (Width, Height):", img.size)
print("Number of Channels:", len(img.getbands()))
print("Channel Names:", img.getbands())
print("Data Type:", arr.dtype)
print("File Size:", os.path.getsize(filename), "bytes")

display(img)
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets.webp
File Name: image for open cv labsheets.webp
Image Format: WEBP
Dimensions (Width, Height): (505, 900)
Number of Channels: 3
Channel Names: ('R', 'G', 'B')
Data Type: uint8
File Size: 40776 bytes
```



Q2. Convert a Colour Image to Grayscale and Display Both

```
# Shiv(55)
from google.colab import files
from PIL import Image
import matplotlib.pyplot as plt

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename).convert("RGB")
gray = img.convert("L")

img.save("original.png")
gray.save("grayscale.png")

plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")

plt.show()

files.download("original.png")
files.download("grayscale.png")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (1).webp
```



Q3. Convert an Image into JPG, PNG and BMP and Compare File Sizes

```
# Shiv(55)
from google.colab import files
from PIL import Image
import os

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename).convert("RGB")

img.save("converted.jpg", quality=95)
img.save("converted.png")
img.save("converted.bmp")

formats = ["converted.jpg", "converted.png", "converted.bmp"]

for file in formats:
    size = os.path.getsize(file)
    print(file, ":", size, "bytes")

for file in formats:
    files.download(file)
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (2).webp
converted.jpg : 143119 bytes
converted.png : 437843 bytes
converted.bmp : 1364454 bytes
```

Q4. Generate a Photo Negative of an Image

```
# Shiv(55)
from google.colab import files
from PIL import Image, ImageOps
import matplotlib.pyplot as plt

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename).convert("RGB")
negative = ImageOps.invert(img)
```

```
negative.save("photo_negative.png")

plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(negative)
plt.title("Photo Negative")
plt.axis("off")

plt.show()

files.download("photo_negative.png")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

Saving image for open cv labsheets.webp to image for open cv labsheets (3).webp

Original Image



Photo Negative



Q5. Batch Image Resizer While Preserving Aspect Ratio

```
# Shiv(55)
from google.colab import files
from PIL import Image
import zipfile
import os
import io

uploaded = files.upload()
filename = next(iter(uploaded))

os.makedirs("resized_images", exist_ok=True)

max_width = 500
max_height = 500
image_list = []

# Check whether the uploaded file is a ZIP or an image
if zipfile.is_zipfile(filename):
    with zipfile.ZipFile(filename, "r") as z:
        for name in z.namelist():
            if name.lower().endswith((".jpg", ".jpeg", ".png", ".bmp", ".webp")):
                image_list.append((os.path.basename(name), z.read(name)))
else:
    image_list.append((filename, uploaded[filename]))

# Resize images while preserving aspect ratio
for name, data in image_list:
    try:
        img = Image.open(io.BytesIO(data)).convert("RGB")
        img.thumbnail((max_width, max_height), Image.Resampling.LANCZOS)

        output_name = os.path.splitext(name)[0] + ".jpg"
        output_path = os.path.join("resized_images", output_name)
```

```
        output_path = os.path.join("resized_images", output_name)

        img.save(output_path, quality=95)
        print("Resized:", name, "->", img.size)

    except Exception as e:
        print("Error processing", name, ":", e)

# Package all resized images into a ZIP file
with zipfile.ZipFile("resized_images.zip", "w") as z:
    for name in os.listdir("resized_images"):
        z.write(os.path.join("resized_images", name), name)

print("Total images resized:", len(os.listdir("resized_images")))
files.download("resized_images.zip")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (5).webp
Resized: image for open cv labsheets (5).webp -> (281, 500)
Total images resized: 1
```

Q6. Crop a User-Specified Region of Interest (ROI)

```
# Shiv(55)
from google.colab import files
from PIL import Image
import matplotlib.pyplot as plt

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename)

display(img)
print("Image dimensions:", img.size)

left = int(input("Enter left coordinate: "))
top = int(input("Enter top coordinate: "))
right = int(input("Enter right coordinate: "))
bottom = int(input("Enter bottom coordinate: "))

if 0 <= left < right <= img.width and 0 <= top < bottom <= img.height:
    cropped = img.crop((left, top, right, bottom))
    cropped.save("cropped_image.png")

    plt.imshow(cropped)
    plt.title("Cropped ROI")
    plt.axis("off")
    plt.show()

    files.download("cropped_image.png")
else:
    print("Invalid coordinates!")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (7).webp
```



```
Image dimensions: (505, 900)
Enter left coordinate: 45
Enter top coordinate: 67
Enter right coordinate: 300
Enter bottom coordinate: 400
```

Cropped ROI



Q7. Split RGB Channels, Reconstruct the Image and Verify Pixel Accuracy

```
# Shiv(55)
```

```
from google.colab import files
from PIL import Image
import numpy as np
import matplotlib.pyplot as plt

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename).convert("RGB")

red, green, blue = img.split()

red.save("red_channel.png")
green.save("green_channel.png")
blue.save("blue_channel.png")

reconstructed = Image.merge("RGB", (red, green, blue))
reconstructed.save("reconstructed.png")

original_array = np.array(img)
reconstructed_array = np.array(reconstructed)

print("Pixel-level accuracy:",
      np.array_equal(original_array, reconstructed_array))

plt.figure(figsize=(12, 6))

plt.subplot(2, 3, 1)
plt.imshow(img)
plt.title("Original")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(red, cmap="gray")
plt.title("Red Channel")
plt.axis("off")

plt.subplot(2, 3, 3)
plt.imshow(green, cmap="gray")
plt.title("Green Channel")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(blue, cmap="gray")
plt.title("Blue Channel")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(reconstructed)
plt.title("Reconstructed")
plt.axis("off")

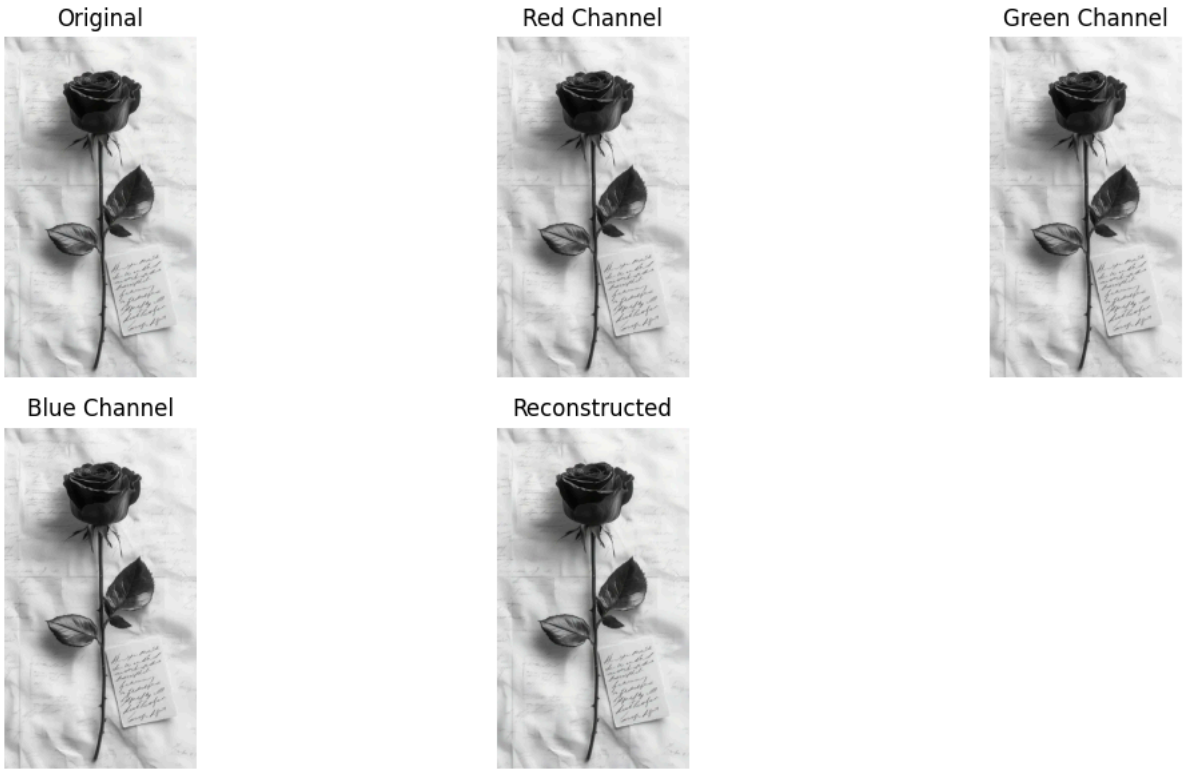
plt.tight_layout()
plt.show()

files.download("reconstructed.png")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (8).webp
Pixel-level accuracy: True
```



Q8. Add a Text Watermark to an Image

```
# Shiv(55)
from google.colab import files
from PIL import Image, ImageDraw, ImageFont

# Step 1: Upload image
uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename).convert("RGBA")

# Step 2: Ask user for watermark text
text = input("Enter the watermark text: ")

# Step 3: Create watermark layer
watermark = Image.new("RGBA", img.size, (0, 0, 0, 0))
draw = ImageDraw.Draw(watermark)

try:
    font = ImageFont.truetype("DejaVuSans.ttf", 35)
except OSError:
    font = ImageFont.load_default()

# Step 4: Calculate text position
bbox = draw.textbbox((0, 0), text, font=font)
text_width = bbox[2] - bbox[0]
text_height = bbox[3] - bbox[1]

x = max(10, img.width - text_width - 20)
y = max(10, img.height - text_height - 20)

# Step 5: Draw background and watermark
padding = 8

draw.rectangle(
    (x - padding, y - padding,
     x + text_width + padding, y + text_height + padding),
    fill=(255, 255, 255, 210)
)

draw.text(
    (x, y), text,
    font=font,
```

```
    fill=(0, 0, 0, 255)
)

# Step 6: Apply watermark
result = Image.alpha_composite(img, watermark)

# Step 7: Save and display result
result.save("watermarked_image.png")

print("Watermark added successfully!")
display(result)

# Step 8: Download output
files.download("watermarked_image.png")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (14).webp
Enter the watermark text: SHIV BHATIA
Watermark added successfully!
```



Q9. Rotate and Flip an Image by User-Defined Angles and Directions

```
# Shiv(55)
from google.colab import files
from PIL import Image
import os
import zipfile

uploaded = files.upload()
filename = next(iter(uploaded))

img = Image.open(filename)
```

```
os.makedirs("transformed_images", exist_ok=True)

angles = input("Enter rotation angles separated by commas (e.g. 90,180,270): ")
angles = [int(a.strip()) for a in angles.split(",")]

for angle in angles:
    rotated = img.rotate(angle, expand=True)
    output = f"transformed_images/rotated_{angle}.png"
    rotated.save(output)
    print("Saved:", output)

horizontal = img.transpose(Image.Transpose.FLIP_LEFT_RIGHT)
vertical = img.transpose(Image.Transpose.FLIP_TOP_BOTTOM)

horizontal.save("transformed_images/flip_horizontal.png")
vertical.save("transformed_images/flip_vertical.png")

with zipfile.ZipFile("transformed_images.zip", "w") as z:
    for name in os.listdir("transformed_images"):
        z.write(os.path.join("transformed_images", name), name)

files.download("transformed_images.zip")
```

Choose Files image for o…sheets.webp

image for open cv labsheets.webp(image/webp) - 40776 bytes, last modified: 10/11/2026 - 100% done

```
Saving image for open cv labsheets.webp to image for open cv labsheets (12).webp
Enter rotation angles separated by commas (e.g. 90,180,270): 90,180,270
Saved: transformed_images/rotated_90.png
```